

IMPLEMENTATION GUIDE

Camera Viewport, Watermarks and Rapid Multi-Image Capture

A separate, simple camera route with the existing upload flow preserved as fallback

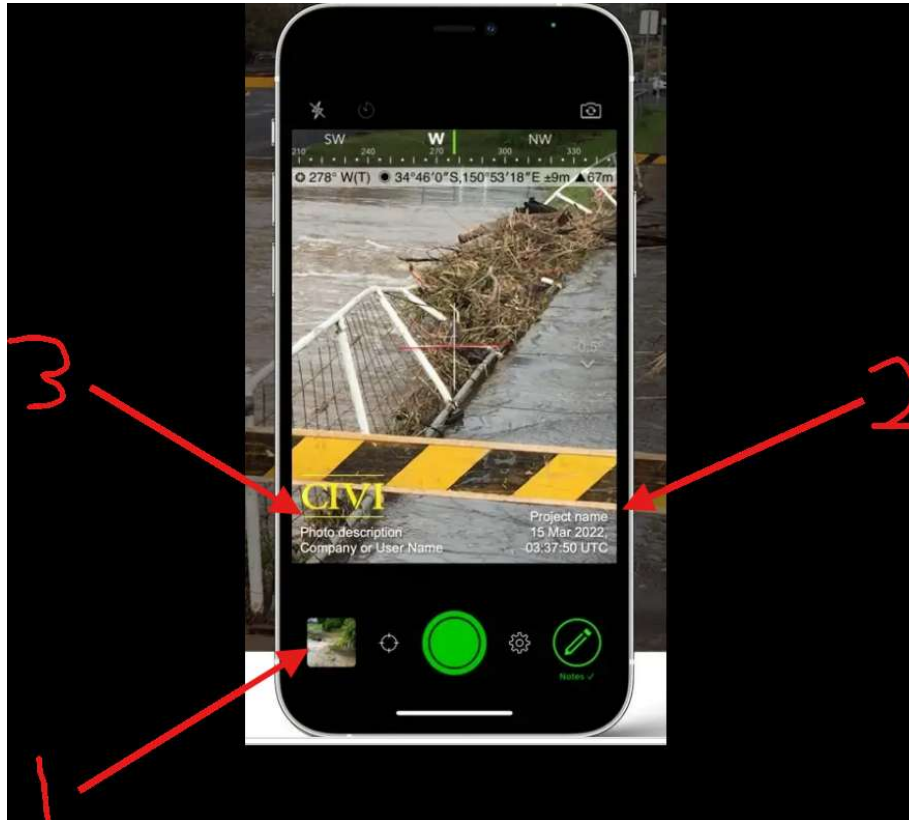


Figure 1. User-supplied reference: gallery access (1), job/date/time values (2), and the Label value (3).

Primary safety rule

The camera works now. Keep its package, permissions, lifecycle, capture function, file model, database, metadata, and existing markup layer intact. Add only a thin viewport and JPEG-composition layer around the proven capture flow.

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1. Objective and design intent

Improve the camera screen without replacing the underlying camera. The finished experience should open directly into a live viewport, support quick repeated photographs, show selected watermark values, and offer a Label lock for uninterrupted field work.

Simple inspiration only

The supplied screenshot is a general layout reference for an open camera viewport, visible text, gallery access, a central shutter, and a reusable Label. It is not a request to reproduce another app or add its broader feature set.

1.1 Required outcome

- Open directly into the existing live camera preview.
- Take multiple images quickly without navigating away after each capture.
- Show a latest-photo thumbnail that opens the existing file viewer.
- Keep the current upload button and its Upload from Camera Roll / Take Photo chooser unchanged as a fallback.
- Display only the enabled watermark values: Label, Date, Time, and Job Number.
- Use a large bottom-centre shutter, with Settings and a Label lock nearby.
- In locked mode, reuse one Label without prompting after every capture.
- In unlocked mode, capture safely first and then request a Label for that photograph.
- Flatten the camera image and visible text watermark into a JPEG while leaving the existing markup layer separate and editable.

1.2 Non-goals

- No camera-library replacement or upgrade.
- No camera permission or lifecycle rewrite.
- No database schema, table, field, relationship, or migration changes.
- No deliberate EXIF or image-metadata changes.
- No change to existing file IDs, storage paths, or established save relationships.
- No change to the existing second markup layer, its coordinates, serialization, storage, or editor.
- No broad refactor and no unrelated screen changes.
- No replacement of the shared upload chooser already used by Forms, Incidents, or other workflows.

1.3 Protected existing upload workflow

The new camera viewport must be implemented as a separate route. The current upload button continues to open its existing Upload from Camera Roll / Take Photo chooser, including its existing Take

Photo behavior. That workflow is already used widely by Forms and Incidents and remains the safe fallback.

- Do not remove, rename, restyle, or bypass the existing upload button unless a local layout wrapper is unavoidable.
- Do not remove or change Upload from Camera Roll.
- Do not change the chooser's current result handling, validation, upload path, or database behavior.
- Do not redirect or modify the chooser's existing Take Photo action.
- Add the new camera viewport as a separate named route with its own entry point.
- Forms, Incidents, and every other current consumer continue using the existing upload workflow unchanged.
- If the new route fails or is unsupported, users can still use the original chooser, Camera Roll, and Take Photo fallback.

1.4 Correction to Airo's expanded research proposal

Airo correctly identified the visible thumbnail and stamped-overlay pattern, but the proposed GPS/metadata feature set is not requested. The following boundaries override that proposal:

- Do not add GPS-coordinate burn-in.
- Do not add compass bearing, altitude, DeviceOrientationEvent handling, KML/KMZ/CSV export, or cloud-storage work.
- Do not introduce dual saving of an original and stamped copy. Continue producing one JPEG through the current working save path unless the existing application already does otherwise.
- Do not change, supplement, or rely on EXIF metadata.
- Do not change the upload queue, server, database, or file relationships.
- Use only the enabled Settings options: Label, Date, Time, and Job Number.
- Locked mode sets/reuses the Label before capture; unlocked mode captures safely first and asks for that photograph's Label afterward.
- The existing second markup layer remains independent and editable.

Scope decision

This is a focused, simple camera-viewport improvement—not a clone of another camera app. GPS, compass, altitude, dual-file archival, exports, and metadata work are explicitly excluded.

2. Required pre-implementation inspection

Before writing code, inspect the working implementation and report the following. Do not assume component names or architecture.

1. The camera-screen component and the component containing the live preview.
2. The existing capture function and its current success/error sequence.

3. The existing JPEG or image-save function and storage destination.
4. How the selected job number reaches the camera screen.
5. Where the current Label can safely live in temporary screen state.
6. The existing file/photo viewer and its navigation entry point.
7. The existing upload button, its Camera Roll / Take Photo chooser, and every shared consumer including Forms and Incidents.
8. The existing second markup layer and how it attaches to the saved base image.
9. The exact files proposed for modification.
10. Risks involving orientation, preview cropping, image resolution, or platform differences.

Checkpoint required

Create a Git checkpoint or recoverable backup before editing. Test the current camera before any change so there is a known-good baseline. Do not start by upgrading dependencies or changing configuration.

2.1 Required implementation response

- Confirm that no database migration is required.
- Confirm that no metadata changes are required.
- Confirm that the current camera library will remain unchanged.
- Describe the proposed full-resolution JPEG-composition method.
- Explain how the markup layer will remain independent.
- List every file to be changed before editing begins.

3. Layered architecture

Treat the feature as a presentation and composition layer around the existing capture pipeline.

```

Camera screen
├── Existing live camera preview
├── New temporary visual text-watermark overlay
├── Existing camera controls
└── Existing markup system remains separate

```

3.1 JPEG composition

```

Captured camera image
      +
Visible text watermark
      =
Flattened JPEG

```

Pass the flattened JPEG through the existing file-saving process. The watermark becomes permanent pixels in the base JPEG; it is not stored as an editable markup and is not written into metadata.

3.2 Existing markup layer

```
Flattened base JPEG (camera + permanent text watermark)
+
Existing editable markup layer
```

- Do not merge existing markups into the camera watermark.
- Do not create watermark markup objects.
- Do not alter markup coordinates, serialization, database records, or editing behavior.
- Do not flatten existing markups during camera capture.
- The new flattened JPEG simply becomes the base image under the unchanged markup system.

3.3 Temporary capture context

At shutter press, create an immutable in-memory capture context so later job changes cannot affect the image being processed.

Use the existing job identifier only where the established save process already requires it. Do not add a new persistent data source.

```
captureContext = { jobNumber, label, capturedAt }
```

4. Camera viewport and bottom controls

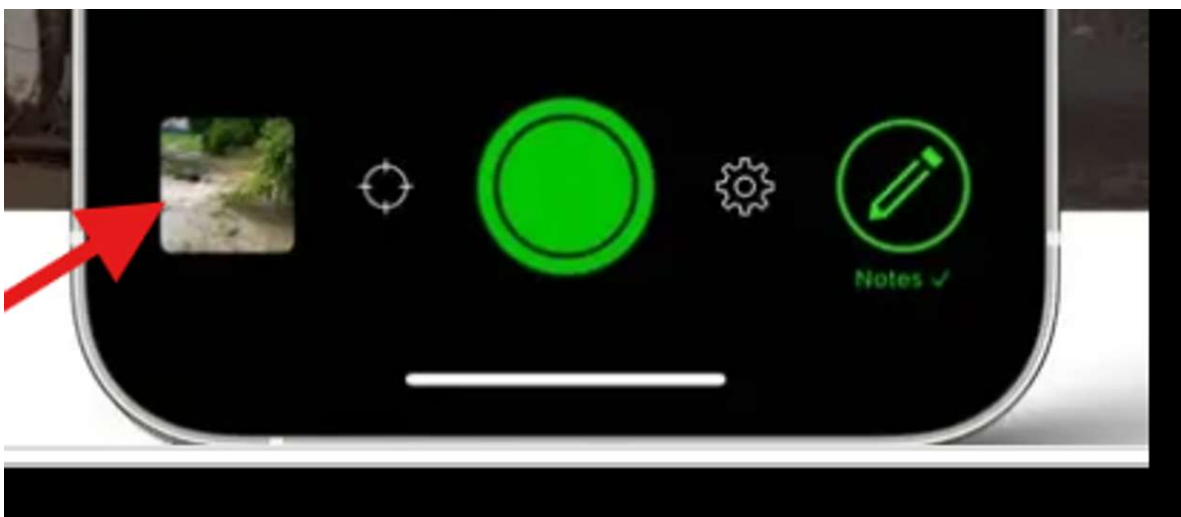


Figure 2. Reference crop showing the lower camera-control area: latest photo on the left and the central shutter.

The camera opens directly into the current live preview. Keep the preview active after successful captures and avoid unnecessary camera reinitialization.

4.1 Control order

[Latest photo] [Settings] [Large shutter] [Label lock]

4.2 Latest-photo button

- Place at bottom left.
- Show the most recently captured photograph; use a gallery/file icon when none exists.
- Open the existing file/photo viewer and reuse its current navigation.
- Do not create a duplicate gallery or change existing file loading.

4.3 Settings

- Reuse the existing settings interface.
- Provide four independent watermark visibility options: Show Label, Show Date, Show Time, and Show Job Number.
- Use the existing settings-storage approach if these preferences can be added without a database change; otherwise keep them in the existing local UI-preference mechanism.
- The live viewport preview and flattened JPEG must use the same enabled/disabled selections.
- If an option is disabled, omit that value completely and close the spacing cleanly.
- Do not duplicate or silently change camera resolution, quality, flash, permissions, or capture configuration.
- Opening and closing Settings must not incorrectly recreate or break the camera.

4.4 Shutter

- Place at bottom centre and make it the largest control.
- One deliberate tap captures one photograph.
- Keep the user on the camera screen and show a brief visual capture acknowledgement.
- Disable only while a capture is actively processing; prevent simultaneous capture operations.
- Re-enable as soon as safely possible for rapid consecutive photographs.
- Never include the shutter or other controls in the saved JPEG.

5. Label lock behavior

The lock controls whether one temporary Label is reused or a new Label is requested for every photograph. Its state must be obvious without relying on colour.

5.1 Locked mode: rapid field capture

- Show a closed-padlock icon and an accessible label: Label locked.
- If no Label exists when locking, open a Label editor.
- Provide a clear action such as Use and lock.
- Hold the Label only in camera-screen state.
- Display it in the live watermark preview.
- Reuse it for every subsequent photograph without a post-capture prompt.
- Allow editing from the camera screen; edits affect future JPEGs only.
- On job change, clear it or explicitly request confirmation before reuse.

Locked-mode intent

This is the fastest workflow. The operator can take several related photographs without interruption while seeing the exact Label that will be stamped.

5.2 Unlocked mode: one Label per image

- Show an open-padlock icon and an accessible label: Label unlocked.
- Capture and safely retain the image first.
- Record the capture timestamp immediately at shutter press.
- Open a small modal or bottom sheet requesting a Label.
- After confirmation, render the watermark, create the JPEG, and use the existing save process.
- Close the editor and return to the still-active camera preview.
- Do not reuse the previous Label automatically.
- Cancellation or dismissal must never corrupt an image or create an incomplete record.

Add photo label

[_____]

[Skip]

[Save]

If the current application requires Labels, preserve that rule and omit Skip. Do not introduce a separate full-screen navigation flow if a safe modal or bottom sheet works.

6. Watermark content and presentation

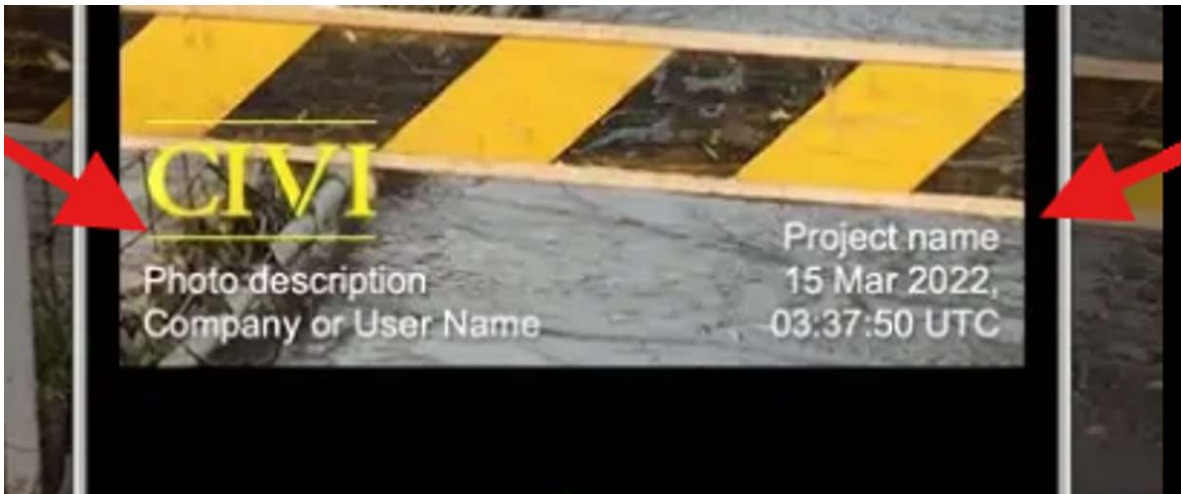


Figure 3. Reference crop showing the watermark area. The new design should use values only and a clearer compact layout.

6.1 Values only

Settings must provide four independent watermark options: Label, Date, Time, and Job Number. Each option can be shown or hidden. The JPEG displays values only—not field names.

```
Damage to northern safety rail
13 Aug 2026
3:37:50 PM
JOB-1234
```

- Normally show each value on its own line.
- Omit an empty optional value cleanly.
- Never show null, undefined, placeholder text, an empty label, or an unnecessary blank row.
- Use the existing job number already available to the camera screen.
- Use the exact timestamp from shutter press, not Label entry or save completion.
- Use the existing application format or the device's local date, time, and locale.

6.2 Appearance

- High-contrast text with a subtle shadow or semi-transparent dark panel.
- Safe spacing from image edges and a readable, compact font.
- Consistent relative placement in portrait and landscape.
- Dynamic height for wrapped Labels.
- Long Labels wrap within bounds, use a sensible maximum line count, and never change the image aspect ratio.
- The preview overlay should closely represent the final saved JPEG.

7. Full-resolution JPEG generation

Preferred method

Use the normal full-resolution output from the existing working capture function, then render the visible watermark onto an off-screen image-sized canvas. This preserves quality and avoids capturing screen controls.

- Obtain the normal full-resolution image from the existing capture function.
- Create an off-screen canvas or equivalent using the image's real pixel dimensions.
- Calculate watermark placement using relative coordinates.
- Render the same values and visual treatment visible in the viewport.
- Export a standard flattened JPEG.
- Pass the JPEG to the existing save function.

7.1 Coordinate mapping

```
watermarkX    = imageWidth  × relativeX  
watermarkY    = imageHeight × relativeY  
watermarkWidth = imageWidth  × relativeWidth
```

Account for preview scaling, cover-cropping, letterboxing, JPEG rotation, portrait/landscape orientation, and front/rear camera behavior if supported. Never stretch or distort the image.

7.2 What must not appear in the JPEG

- Shutter, Settings, lock, or latest-photo controls.
- Navigation bars, safe-area padding, dialogs, bottom sheets, or status messages.
- Focus or exposure indicators and loading states.
- The existing editable markup layer.

7.3 Screenshot clarification

The result should look like a clean capture of the bounded camera viewport, but do not take a literal full-screen screenshot. A bounded viewport capture is acceptable only if the current framework already supports it safely, excludes controls, and preserves adequate resolution. Do not add a risky screenshot dependency.

8. File, metadata, and database safety

8.1 File safety

- Do not overwrite existing photographs.
- Continue using the existing unique filename, path, or identifier system.
- Do not change the storage directory.
- Do not create incomplete file/database records before the JPEG is ready unless the existing proven pipeline already does so.
- Keep the captured source safely in temporary processing state until composition succeeds.

8.2 No metadata changes

- Do not add watermark values to EXIF.
- Do not deliberately rewrite or supplement metadata.
- Do not alter the existing database timestamp; the capture time is visual text only.
- Do not introduce a new metadata-processing package.

8.3 No database changes

- No tables, columns, schemas, migrations, or relationship changes.
- Do not persist the Label lock state in the database.
- Do not add a second Label or watermark record.
- Do not change existing file IDs or job/file relationships.

8.4 Failure handling

- Never overwrite another file or report a failed composition as successfully saved.
- Show a clear error and allow retry or safe return to the camera.
- Do not create a broken database record.
- Preserve the existing rollback/error-handling behavior when one already exists.

9. Incremental implementation sequence

Stage 1 — Inspect and checkpoint

Identify relevant components and functions, report proposed files, create a checkpoint, and verify the current camera.

Stage 2 — Control layout only

Add or rearrange bottom controls without changing capture behavior; retest preview and capture.

Stage 3 — Existing viewer connection

Connect the latest thumbnail to the existing viewer and verify safe return to the camera.

Stage 4 — Temporary lock state

Add in-memory locked/unlocked state and the Label editor; do not touch database persistence.

Stage 5 — Visual preview overlay

Show only the Settings-enabled Label, Date, Time, and Job Number values; test portrait and landscape without saving it yet.

Stage 6 — Flattened JPEG

Wrap the existing capture result, render the watermark, and use the existing save process.

Stage 7 — Locked workflow

Reuse the temporary Label and test rapid multi-image capture without prompts.

Stage 8 — Unlocked workflow

Capture first, prompt for a Label, compose, and save; verify cancellation safety.

Stage 9 — Regression testing

Retest permissions, capture, save, viewer, job association, markup editing, orientation, errors, reopening, and job switching.

Stop rule

If any stage breaks existing camera behavior, stop. Restore the checkpoint or revert only that stage before proceeding. Do not compensate with a broad rewrite.

10. Acceptance checklist

- ☐ The camera opens directly into the live preview.
- ☐ Existing camera permissions still work.
- ☐ The centre shutter produces one photograph per tap.
- ☐ The user remains on the camera screen after capture.
- ☐ Five photographs can be taken consecutively in locked mode without freezing.
- ☐ The camera does not unnecessarily reinitialize between photographs.
- ☐ The latest-photo thumbnail updates and opens the existing viewer.
- ☐ Settings still work and safely return to the camera.
- ☐ Locked and unlocked modes have clear icons and accessible labels.
- ☐ Locked mode reuses one Label without prompting.
- ☐ Unlocked mode requests a Label for each photograph.
- ☐ Changing a locked Label affects future JPEGs only.
- ☐ Switching jobs cannot silently reuse the previous job's Label.
- ☐ The JPEG contains the correct enabled values among Label, Date, Time, and Job Number.
- ☐ Each of the four Settings options independently shows or hides the matching value.

- ☐ The watermark contains values only, with no field labels.
- ☐ Empty optional values do not create broken rows or placeholder text.
- ☐ The watermark is readable in portrait and landscape.
- ☐ The JPEG contains no camera controls, dialogs, or editable markups.
- ☐ The image is not stretched, incorrectly cropped, or incorrectly rotated.
- ☐ The output remains a valid JPEG at acceptable quality.
- ☐ No deliberate metadata changes or additions occur.
- ☐ No database migration or schema change is created.
- ☐ Existing file IDs, paths, and relationships remain valid.
- ☐ Existing files are not overwritten.
- ☐ Existing markups still load and remain separate and editable.
- ☐ The text watermark is not created as an editable markup object.
- ☐ A composition failure cannot corrupt or falsely save an image.
- ☐ Previously saved images remain unchanged.
- ☐ The existing upload button still opens the same Camera Roll / Take Photo chooser.
- ☐ Upload from Camera Roll still works unchanged.
- ☐ The new viewport opens through a separate route and does not intercept the chooser's original Take Photo action.
- ☐ Forms, Incidents, and other existing consumers continue to work without regressions.

11. Copy-ready instruction to Airo

Implementation authority

Use this section as the controlling instruction. Existing working behavior and the restrictions in this guide take priority over aesthetic convenience.

Use the supplied screenshot only as simple layout inspiration. Build a camera that remains open, with a lower-left existing photo-viewer thumbnail, a large central shutter, the Settings-enabled Label/Date/Time/Job Number values, and a Label lock. Do not reproduce another app's branding, visual design, or broader feature set.

The existing camera is working and must remain intact. Do not replace the camera library, capture function, permission handling, lifecycle management, file model, database, metadata, or existing markup system. Implement the requested behavior as a thin UI and in-memory state layer around the existing capture function.

Implement the new camera viewport as a separate route with a separate entry point. Keep the current upload button and its Upload from Camera Roll / Take Photo chooser exactly functional as the protected

fallback, including its original Take Photo action. It is already shared by Forms, Incidents, and other workflows. Do not change its routes, result contract, validation, uploads, or shared consumer behavior.

The app already has a separate second markup layer. Preserve it exactly. The camera watermark must be permanently flattened into the new base JPEG, while the existing markup layer remains separate and editable above that JPEG.

Build a camera viewport with the existing live preview as the bottom layer and a temporary text-watermark overlay above it. When the shutter is pressed, use the working full-resolution camera output and render the visible watermark onto a JPEG-sized off-screen canvas. Pass the resulting flattened JPEG through the existing save process. Do not take a literal full-screen screenshot, capture controls, change EXIF, or write new watermark information into the database.

Settings must contain four watermark options: Label, Date, Time, and Job Number. The watermark displays only enabled values and never displays field names. Use the timestamp recorded at shutter press and omit disabled or empty values cleanly.

Locked mode reuses one temporary Label for rapid captures without prompting. Unlocked mode captures and retains the image first, then asks for a Label for that photograph before composition and saving. Do not allow cancellation to corrupt an image or create an incomplete record.

Before editing, create a Git checkpoint and report the exact components, functions, and files involved. Confirm that no database migration, metadata change, camera-library replacement, or markup-layer change is required. Implement one stage at a time and retest the working camera after every stage.

12. Visual reference

The screenshots in this guide are derived from the user-supplied annotated reference image. They communicate only the desired general layout and interaction pattern.